

Image Processing – HW2

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Please refer to the README.md first, the following only explain the code logic and the reason for the result.

Q1:

Steps:

1. Make it binary
2. Closing operation to get the repaired image
3. Apply Dilation to that repaired image then subtract afterwards to get outer boundary

Reason:

It's just basic OpenCV function call with basic morphological operations, all as expected. For images, please see README.md to try to locate.

Q2:

Steps:

1. Stretch the local min, max to 0 and 255 linearly

Reason:

```
Original min: 0, Original max: 255  
---Stretch to---  
Global min: 0, Global max: 255
```

As you can see, there's no difference between the contrast stretched version and the original one because the local min and max for the original image is 0 and 255. Applying linear contrast stretching will do absolutely nothing.

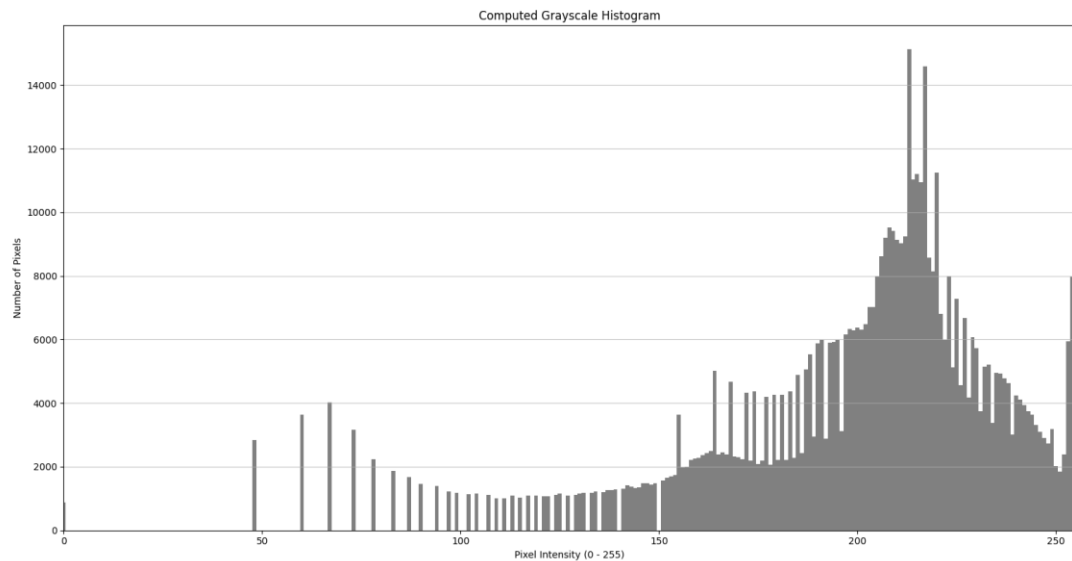
Q3:

Steps:

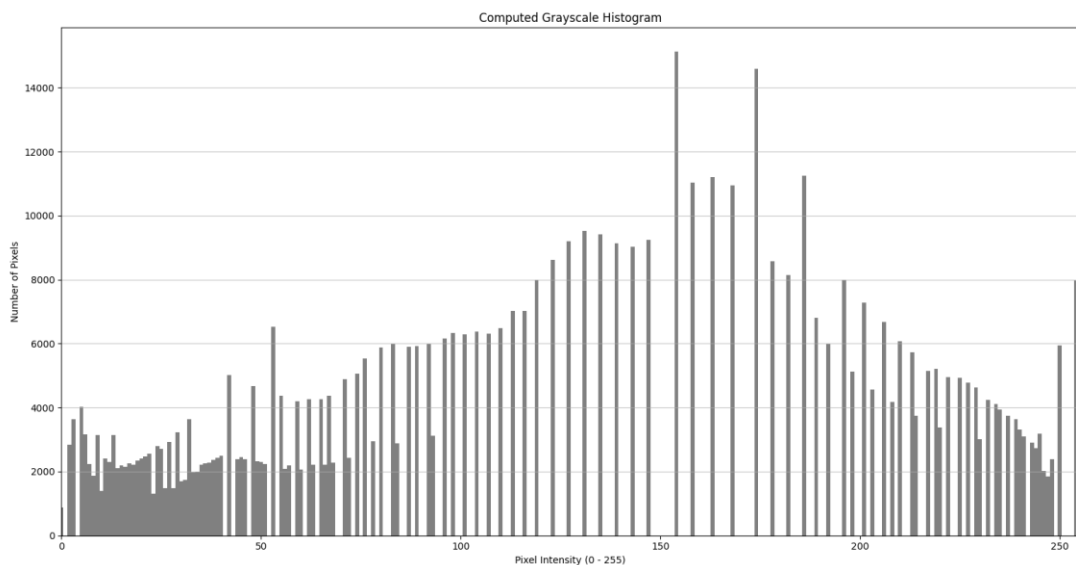
1. Generate the histogram from input images
2. Sum the probability to get CDF
3. Creating an intensity lookup table
4. Apply mapping (histogram specification)

Reason:

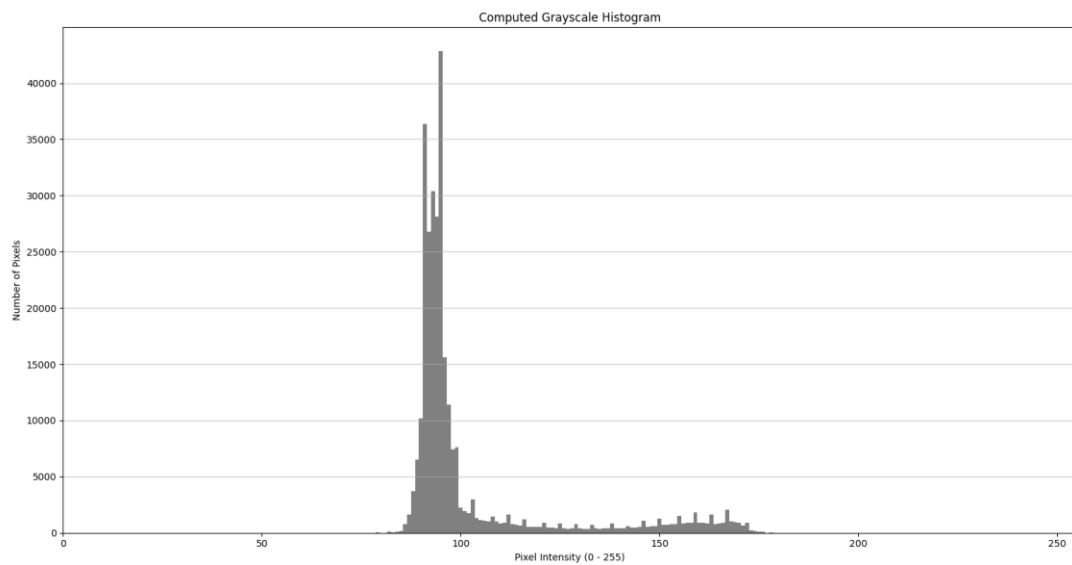
For aerialview-washedout.tif, it's working very well because



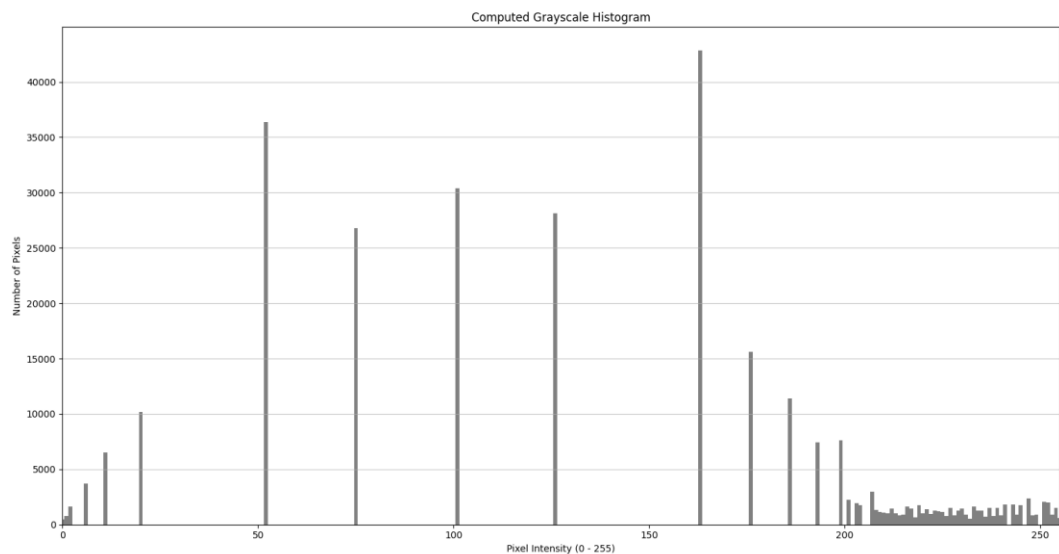
As you can see, the overall intensities are pretty spread out. Therefore, it can be mapped effectively without a particular peak.



For einstein-low-contrast.tif, it's the opposite. Take a look at the original histogram:



Almost all pixels are in 90 ~ 100. When we GHE this, the image will be overly bright just to match this heavily skewed case.



As you can see, lots of pixels are around 200 ~ 250 just to satisfy the CDF.

Q4:

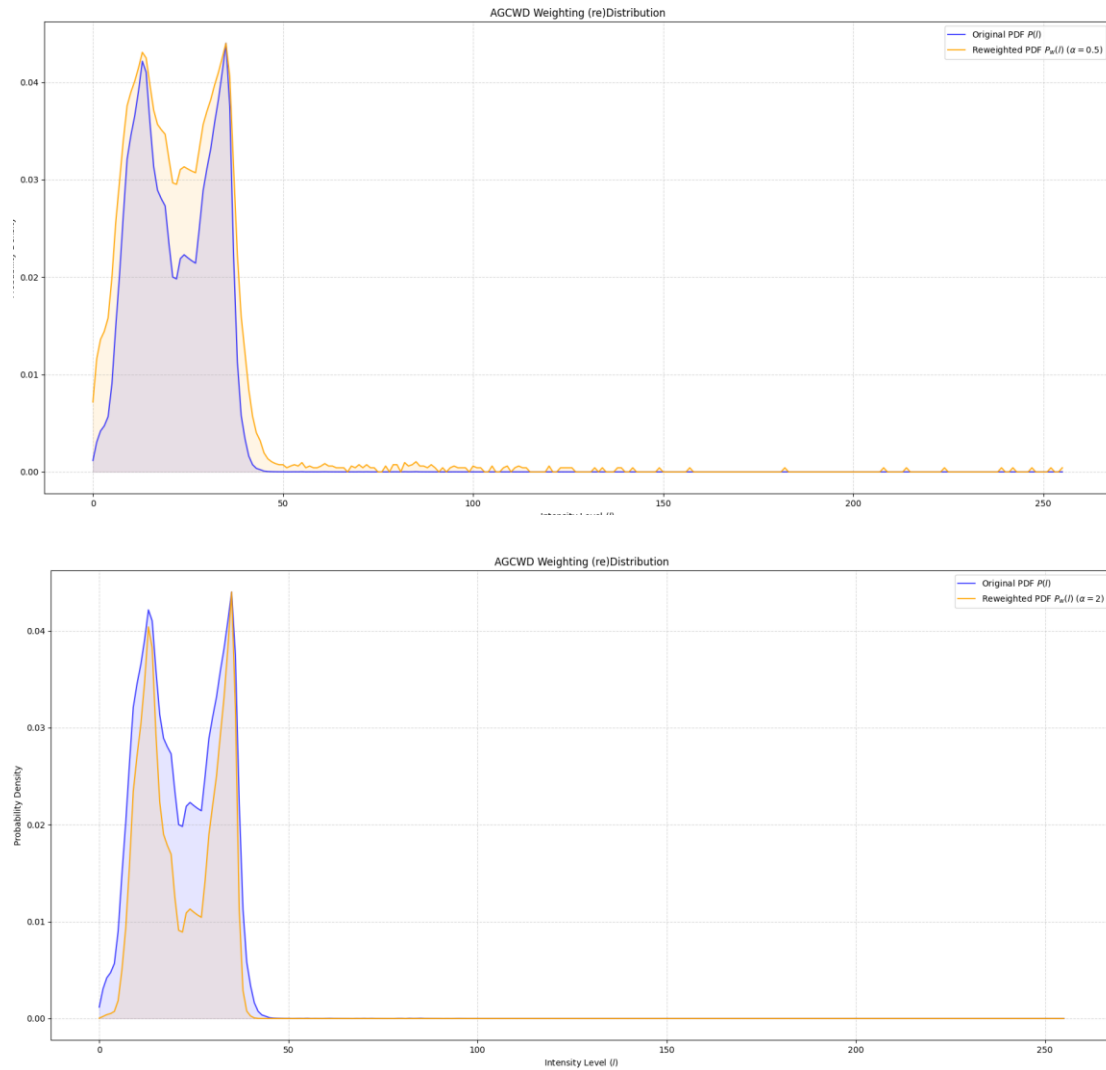
Steps:

1. Generate the histogram from input images

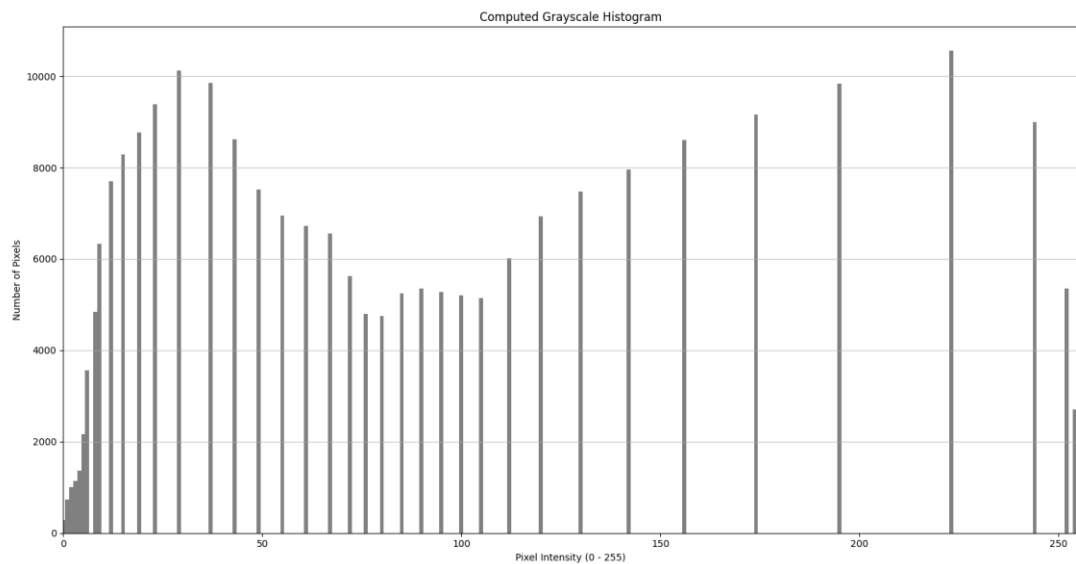
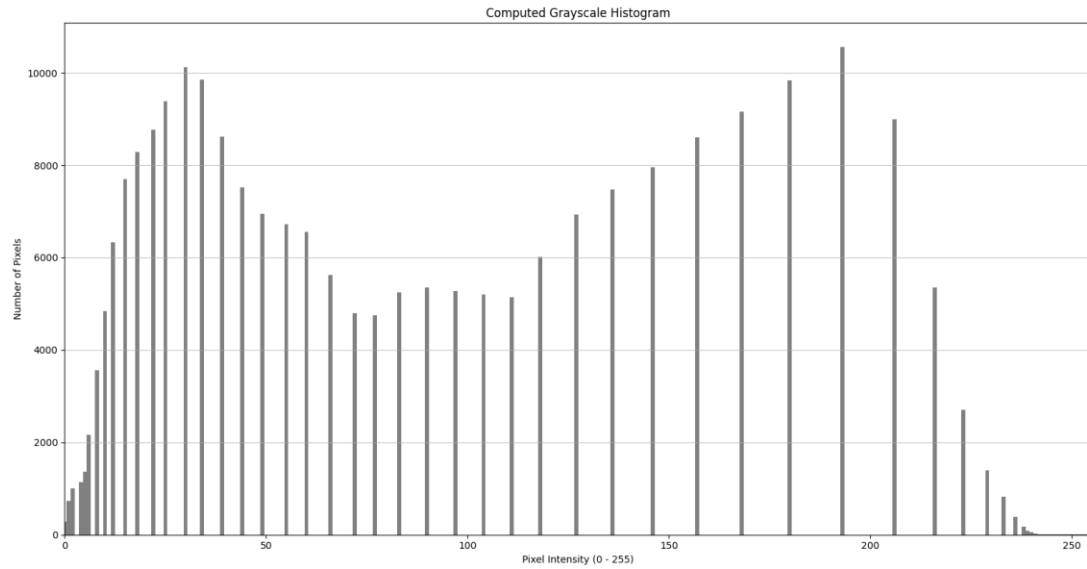
2. Re-weight using alpha
3. Calculate the CPF_w
4. Apply Gamma correction for each intensity

Reason:

It seems weird that the 2 images with different alpha (0.5 and 2) looks almost the same, but here's the breakdown:



As you can see, the re-weighted PDF only makes a difference in intensities around 25 ~ 30, which is not that obvious for human eyes to differentiate. Look at the AGCWD images' histogram too:



Not really much of a difference.

Q5:

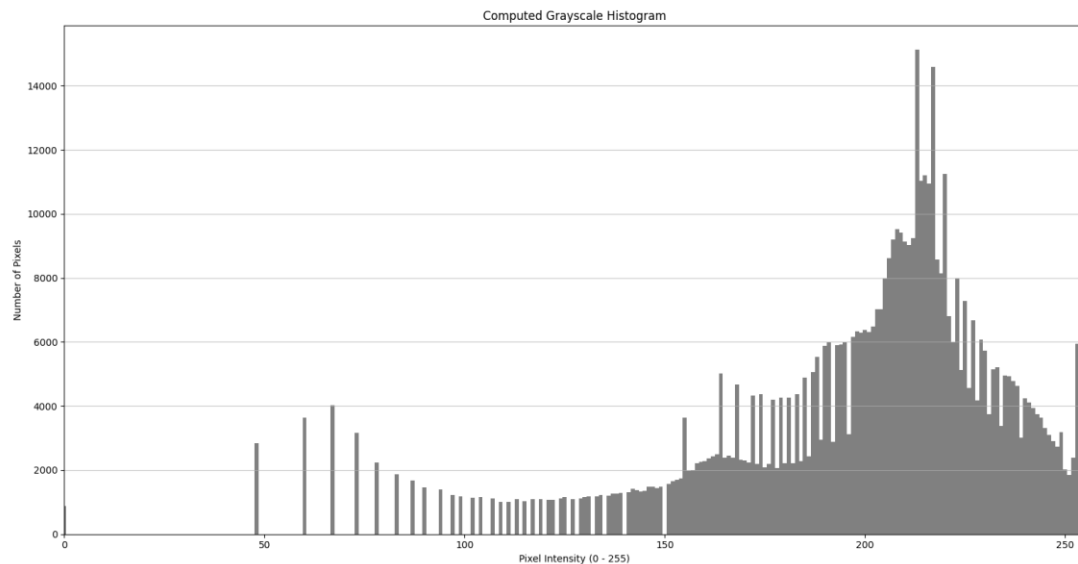
Steps:

1. Get mean, median intensity
2. Just like Q3 but we set the start and end intensity zone
3. Since the 2 partitions won't overlap, we can just get 2 images using 2 HE and add them together

Reason:

Mean: 197
Median: 208

Since the mean and median are quite similar, we can discuss them together (use Median as example). The original image's histogram looks like:



Pixels from 0 ~ 197 and 198 ~ 255 are all mapped to 0 ~ 255, which means there are pixels that have different intensity but mapped to the same intensity after HE. This creates a feeling of incongruity.

Q6:

Steps:

1. Construct 2D Histogram using np's advanced indexing
 - Especially in applying kernels to the image, instead of padding, we set boundary mask so that center_vals and neighbor_vals are talking to each other conveniently.
2. Normalization (in 2D)
3. Compress the 2D normalized histogram to a 1D CDF array
4. Creating an intensity lookup table
5. Apply mapping (histogram specification)

Reason:

It's working pretty well.



It looks like spatial information is preserved because the components of the image are obviously more natural than the one in question 3's GHE.

